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Spatial Patterns and Determinants of Multidimensional Beta Diversity in the United States Freshwater Fishes

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ABSTRACT

Aim: Understanding the spatial patterns and drivers of biodiversity is foundational to ecology and biogeography and crucial for effective conservation. Given the critical ecological role of freshwater fishes, large-scale studies examining the multiple dimensions of their beta diversity are urgently needed.

Location: The United States.

Methods: We analysed the beta diversity of 786 freshwater fish species across 1873 watersheds in the US by integrating their distribution, functional traits and phylogeny. Taxonomic, functional and phylogenetic beta diversity were deconstructed into nestedness and turnover components. Boosted Regression Tree (BRT) models then quantified the partial dependency of contemporary, historical climate and geographical environmental factors on each dimension.

Results: Our findings revealed that the geographical distribution patterns of the three dimensions of beta diversity, including their nestedness and turnover components, exhibited substantial congruence across the US freshwater fish communities. Taxonomic beta diversity (average 0.43 ± 0.13) was primarily driven by turnover (0.25 ± 0.13), whereas nestedness contributed more strongly to functional beta diversity (0.34 ± 0.16; total beta diversity $= 0.52 \pm 0.19$) and phylogenetic beta diversity (0.17 ± 0.09; total beta diversity $= 0.30 \pm 0.11$). Contemporary precipitation and climate anomalies played significant roles in shaping beta diversity. Late Quaternary glacial–interglacial climate-change velocity, elevation, runoff and geographical distance emerged as the primary explanatory variables for the beta diversity patterns.

Main Conclusions: This study examines the dominant patterns and drivers of multidimensional beta diversity in US freshwater fishes. Functional beta diversity exceeded taxonomic and phylogenetic dimensions. Taxonomic diversity was driven mainly by species turnover, while functional and phylogenetic diversity were dominated by nestedness, with all three showing significant spatial correlations. These patterns were primarily shaped by Late Quaternary climate-change velocity, elevation, runoff and geographic distance. We recommend prioritising conservation in central and western US watersheds with strategies tailored to turnover- and nestedness-dominated regions.

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1 | Introduction

Understanding spatial variation in biodiversity is a crucial aspect of ecology and biogeography, providing valuable insights for biodiversity conservation. Biodiversity is typically quantified across three hierarchical levels: alpha diversity (within a single community), beta diversity (between communities) and gamma diversity (at the regional scale) (Whittaker 1960). Among these, beta diversity serves as a powerful indicator of compositional differences between communities, moving beyond species richness to capture nestedness and turnover patterns that reflect ecological and evolutionary processes (Baselga 2010; Socolar et al. 2016). Beta diversity can be dissected into two components: nestedness (βjne) and turnover (βjtu). Both total beta diversity and its individual components could be calculated using Jaccard or Sørensen dissimilarity indices (Baselga 2010; Baselga et al. 2012). Jaccard and Sørensen indices provide an intuitive measure of dissimilarity between communities and are widely used in biodiversity research (Moreno et al. 2018). Nestedness represents the degree to which the species in a locality with fewer species are a subset of the species in the locality with more species. On the other hand, turnover refers to beta diversity attributed to species replacement and may reflect species sorting due to different environmental conditions or dispersal processes (Baselga 2010; Baselga et al. 2012). An assemblage can either be a strict subset of another, demonstrating full nestedness or it can exhibit the same species richness while having entirely different species compositions, indicating full turnover (Baselga 2010).

Taxonomic beta diversity (TDβ) is a traditional measure widely used to characterise changes in species composition across space and time (Swenson et al. 2012). However, TDβ alone cannot account for differences in species' functional traits or phylogenetic relationships, which limits its ability to reveal the underlying mechanisms structuring communities (Swenson et al. 2012). Incorporating taxonomic, functional traits and evolutionary history dimensions into beta diversity studies can reveal the integrated ecological mechanisms structuring communities (Roa-Fuentes et al. 2019). Functional beta diversity (FDβ) measures the value and range of functional traits among different species assemblages (Mason et al. 2013). FDβ information quantitatively assesses niche differences between species, emphasising functional distinctions among them (Laughlin 2014; Su et al. 2019, 2023). Phylogenetic beta diversity (PDβ) captures evolutionary disparities, allowing comparisons even between communities with no species in common (Qian et al. 2021). Although TDβ, FDβ and PDβ are often correlated, they may decouple. For example, high taxonomic turnover coincides with low functional differentiation due to environmental convergence (Villéger et al. 2013). Integrating these three dimensions thus provides a more mechanistic understanding of the factors structuring biodiversity.

The concurrent examination of TDβ, FDβ and PDβ has been employed in several studies, showcasing its utility in providing a more comprehensive understanding of ecological patterns (Teichert et al. 2018; García-Girón et al. 2020; Xu et al. 2023; Chen et al. 2025). For example, Roa-Fuentes et al. (2019) demonstrated that integrating these three dimensions in stream fishes revealed complementary insights into community assembly mechanisms. More recently, global-scale analyses have further highlighted the value of multidimensional approaches for understanding freshwater fish diversity patterns and their drivers (Chen et al. 2025). Despite these insightful studies, it is notable that most research still focuses on one or two indices. This gap is particularly pronounced in freshwater fishes. Despite comprising one-quarter of all vertebrate species and playing irreplaceable functional and provisioning roles in aquatic ecosystems (Dudgeon et al. 2006; Tedesco et al. 2017), the spatial drivers of their community composition remain understudied. A multifaceted approach that integrates TDβ, FDβ and PDβ perspectives would therefore provide a more mechanistic understanding of freshwater fish community assembly and directly inform conservation strategies for these ecologically and economically vital organisms (Pool et al. 2014; Roa-Fuentes et al. 2019; Jia et al. 2020). Biodiversity patterns are shaped by the interplay of historical legacies and contemporary climatic factors, including post-glacial climate shifts, temperature and precipitation regimes, energy gradients and geographic constraints (Baselga et al. 2012; Chen et al. 2025). These drivers have been widely used to explain spatial variation in biodiversity across different systems, illustrating how past and present processes jointly structure biological assemblages (Ordonez and Svenning 2017; Su et al. 2022). Deciphering such complex spatiotemporal influences is therefore essential for advancing both ecological understanding and effective biodiversity conservation.

The US harbours one of the world's highest diversities of post-Pleistocene freshwater fish species among temperate regions (Knouft and Page 2011). Existing biodiversity assessments in this area have primarily focused on taxonomic and phylogenetic beta diversity (Qian et al. 2021). However, a significant gap remains in comprehensively integrating TDβ, FDβ and PDβ—along with their respective nestedness and turnover components—into a unified analytical framework. Such an integrated perspective is especially critical for advancing our understanding of biodiversity assembly mechanisms. Therefore, adopting a unified TDβ-FDβ-PDβ approach is essential for elucidating the processes that drive beta diversity patterns in freshwater fish communities at a continental scale.

In this study, we mapped and analysed the geographic patterns of beta diversity in the US freshwater fish communities and examined their relationships with key environmental factors. We specifically tested three hypotheses: (1) Nestedness constitute a major component of beta diversity. Previous global analyses of freshwater fishes have shown that nestedness dominates functional and phylogenetic beta diversity at broad spatial scales (Chen et al. 2025). This pattern likely emerges because environmental filtering and dispersal limitation impose non-random constraints on species distributions, such that species-poor assemblages tend to represent predictable subsets of species-rich communities rather than random samples (Taylor and Warren Jr 2001); (2) taxonomic, functional and phylogenetic beta diversity exhibit significant spatial congruence, as species' ecological functions and evolutionary histories are generally linked to their taxonomic identities (Pool et al. 2014); (3) both contemporary climate and historical climatic change act as primary drivers of beta diversity patterns.

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Present-day climatic conditions, such as temperature and precipitation, directly shape fish community structure, whereas historical climate influences beta diversity through long-term evolutionary and biogeographic legacies (Leprieur et al. 2011; Sandel et al. 2011). Through this work, we aim to characterise the biodiversity structure of freshwater fish communities across the US, identify the principal factors driving beta diversity and provide ecological insights to support conservation strategies under ongoing environmental change.

2 | Materials and Methods

2.1 | Freshwater Fishes Occurrence Data

This study employs the most comprehensive watershed-scale (HUC8) database of native freshwater fishes distributions across the US, originally compiled by Su et al. (2023) based on species occurrence records sourced from NatureServe (https://www.natureserve.org). To reflect historical natural distributions, all non-native species were excluded and only native species were retained. The resulting dataset includes 786 native freshwater fish species distributed across 1873 watersheds. This represents approximately 91% of the total species pool and 90% of all watersheds, thereby providing a solid spatial foundation for analysing and mapping beta diversity patterns. This meticulously curated dataset provides a reliable representation of native fish distributions across the US.

2.2 | Functional Trait and Phylogeny

The functional trait database used in this analysis was obtained from Su et al. (2023), which compiled 20 morphological traits for freshwater fish species present in the occurrence records. Trait data were assembled from multiple sources: the FISHMORPH database (Brosse et al. 2021), the Fish Traits database for the US freshwater fishes (Frimpong and Angermeier 2009) and FishBase (Froese and Pauly 2018). These traits were divided into two functional categories: ten traits associated with locomotion and food acquisition and another ten related to life-history and physiological functions. The locomotion and food acquisition traits include: maximum body length, body elongation, relative eye size, oral gape position, relative maxillary length, vertical eye position, body lateral shape, pectoral fin vertical position, pectoral fin size and caudal peduncle throttling. The life-history and physiological traits comprise: longevity, fecundity, mature age, trophic level, temperature range, minimum temperature, maximum temperature, euryhaline, parental care and diet breadth. For detailed sourcing of each trait, refer to Su et al. (2023). To quantify functional dissimilarity among species within communities, Gower's pairwise distance method was applied to the 20 traits. This approach accommodates mixed data types, including categorical, ordinal and continuous variables. All traits were equally weighted and scaled following the standard Gower formulation. Principal coordinates analysis (PCoA) was then performed on the resulting distance matrix. To strike a balance between dimensionality reduction and retention of explanatory power, we retained the first four PCoA axes, which together explained more than 70% of the total functional variation (Su et al. 2023). Since all watersheds met the criterion of sufficient species richness (≥ 5 species), no watersheds were excluded from the analytical dataset.

The phylogenetic tree used in this study was derived from the fish phylogenetic framework established by Rabosky et al. (2018), which comprises 31,526 marine and freshwater ray-finned fish species. Among them, the phylogenetic positions of 11,638 species were precisely estimated based on genetic data; to improve species coverage, the remaining 19,888 species were integrated into the phylogenetic tree using the random polytomy resolution method described in the same study. In our analysis, we first extracted a subtree from the Rabosky phylogeny that included all 786 US species. Based on this subtree, we then calculated the phylogenetic distance matrix among species.

2.3 | Data Analysis

2.3.1 | Beta Diversity Calculations

Multiple dimensions of fishes total beta diversity, based on the Jaccard dissimilarity index (βjac), were thoroughly assessed and further deconstructed into nestedness (βjne) and turnover (βjtu) components. The nestedness component was calculated as the difference between total beta diversity and the turnover component, representing the contribution due to differences in species richness, phylogenetic and functional diversity when species-poor assemblages are nested within species-rich assemblages (Baselga 2010). On the other hand, the turnover component reflects the replacement of species, functional traits or phylogenetic lineages, without being affected by differences in functional diversity, species richness or phylogenetic diversity among sites. For pairwise comparisons, it is calculated as the minimum number of unique species, functional traits or phylogenetic lineages at either site divided by the total number of species, functional traits or lineages present across both sites (Baselga 2010). All beta diversity calculations were conducted using the package 'betapart' in R version 4.4 (R Core Team 2020). To standardise the analytical approach and control for potential bias due to variation in the number of adjacent watersheds—particularly since watersheds on islands and along continental margins often have fewer adjacent watersheds—we consistently selected the five spatially nearest neighbouring watersheds for each target watershed in all analyses. Based on pairwise comparisons between each target watershed and its five nearest neighbours, we calculated dissimilarity metrics—including βjac and its nestedness (βjne) and turnover (βjtu) components—for each pair. The mean value of each dissimilarity metric across the five neighbouring pairs was then used as the final measure for the target watershed, which was subsequently applied in further analysis and visualisation.

2.3.2 | Environmental Drivers

To assess the drivers of freshwater fish beta diversity, we initially compiled 23 predictor variables representing climatic and geographic distance dimensions (Table S1). These

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variables were grouped into four major categories: climate anomalies (historical stability), contemporary temperature, contemporary precipitation and geographic factors. To account for spatial autocorrelation, we calculated the spatial proximity for each watershed as the mean Haversine distance to its five nearest neighbours (geographic distance) based on centroid coordinates (Robusto 1957), resulting in an initial pool of 24 predictors.

Variable selection was guided by ecological relevance and well-documented effects on freshwater fish beta diversity. The selected predictors span key dimensions of historical climate, contemporary climate and geography, thereby enabling a comprehensive and systematic assessment of the multiple drivers underlying beta diversity variation across watersheds. To align analytically with beta diversity, which quantifies compositional dissimilarity among sites, we derived a watershed-scale metric of environmental heterogeneity for each predictor. Specifically, for each focal watershed, we calculated the mean absolute difference in each environmental variable between that watershed and each of its five nearest neighbours. The mean of these five absolute differences was then used as an indicator of spatial heterogeneity for that variable within the focal watershed.

To address potential multicollinearity and ecological redundancy among predictors, we implemented a two-step variable selection procedure. First, we examined pairwise correlations among all 24 candidate predictors using Spearman's rank correlation coefficient. Pairs of variables with $|R| > 0.6$ were considered to exhibit high collinearity. Second, when high collinearity was detected, we retained the variable with greater ecological relevance to freshwater fish beta diversity based on theoretical considerations and empirical evidence, while excluding the redundant variable. For example, contemporary temperature variables (such as annual mean temperature and mean temperature of the warmest quarter) showed strong correlations with elevation (Spearman's $|R| > 0.7$). However, given that elevation integrates multiple environmental filters (including temperature) that affect fish distributions, we retained elevation. Through this approach, the selected environmental variables both avoided multicollinearity and maintained ecological interpretability.

From an initial set of 24 predictors, a total of nine were removed due to strong correlation or informational redundancy with other variables (Figure S1). The remaining 15 variables are: watershed area (Area), Late Quaternary glacial–interglacial climate-change velocity (Velocity), Elevation, Runoff, change in mean annual temperature and precipitation between present and Last Glacial Maximum conditions (LGMT and LGMP), temperature seasonality (TSea), precipitation seasonality (PSea), mean temperature of wettest quarter (TWet), mean temperature of driest quarter (TDri), mean temperature of coldest quarter (TCol), precipitation of wettest quarter (PWet), precipitation of driest quarter (PDri), precipitation of warmest quarter (PWar) and mean geographic distance (Distance). This deliberate variable selection was designed to enhance model accuracy and interpretability through the reduction of predictor collinearity. As a result, the refined variable set is better suited to capture the relative contribution of each predictor in shaping TDβ, FDβ and PDβ diversity patterns within freshwater fish communities.

2.3.3 | Statistical Analysis

To examine the congruence among fishes TDβ, FDβ and PDβ, along with their components (nestedness and turnover), Spearman correlations were performed. This analysis aimed to unveil potential relationships and similarities among these different facets of beta diversity within freshwater fish communities.

In the driver analysis, we employed the Boosted Regression Tree (BRT) model (Elith et al. 2008). BRT is an ensemble method that integrates regression trees with a boosting algorithm, constructing relationships between predictors and the response variable through recursive binary splitting (Elith et al. 2008). The model sequentially fits multiple simple regression trees in an additive manner, gradually improving predictive performance by iteratively reducing residuals. This approach requires no complex data preprocessing, accommodates missing values and effectively captures nonlinear relationships and interaction effects, thereby overcoming the limitations of single-tree models in prediction accuracy (Elith et al. 2008; Yahaya et al. 2019). We used beta diversity as the response variable and employed a BRT model to identify key drivers shaping diversity distribution patterns. Additionally, geographic distance was incorporated as a geographic indicator of spatial autocorrelation to examine the influence of spatial dependence on diversity structure. To facilitate clearer visualisation of their relationships in the figures, we applied a log-transformation to the area and elevation variables due to their wide value ranges.

This model relies on four crucial parameters: Learning Rate (lr), which determines the contribution of each tree to the growing model; Bag Fraction (bf), representing the proportion of samples used at each step in model building; Tree Complexity (tc), controlling the fitting of interactions between predictors; and Number of Trees (nt), indicating the boosting iterations for optimal prediction. A smaller learning rate enhances model robustness but requires more trees, while higher bag fractions and tree complexities influence the model's construction. To fine-tune the BRT model, a two-step optimisation process was employed. Firstly, involved 10-fold cross-validation (CV) to screen for optimal values of the four parameters. The goal was to ensure model calibration and effective generalisation to new data. The model with the highest CV-D^2 (deviance explained) was retained as optimal. Secondly, model evaluation aimed at enhancing robustness by running the model 100 times under different random seeds (Su et al. 2023). Mean values of relative predictor influence and the proportion of deviance explained (D^2) were calculated based on these runs. BRT models were fitted utilising the 'gbm.step' function within the 'dismo' package of the R software (Elith et al. 2008). In this study, partial dependence plots were constructed using centred fitted function values (i.e., effect sizes) to illustrate the impacts of individual explanatory variables on beta diversity.

All statistical analyses and data visualisation processes were performed using R software version 4.4 (R Core Team 2020).

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3 | Results

3.1 | Three Facets of Beta Diversity and Their Components

By assessing freshwater fishes beta diversity across US by examining three facets of beta diversity, most watersheds exhibited diversity values between 0.25 and 0.75, with only a few watersheds displaying the highest or lowest values for TDβjac (Figure 1a). TDβjac exhibited a relatively even distribution among the studied watersheds (Figure 1a). Decomposition of TDβjac (mean: 0.43 ± 0.13) into its components revealed that the TDβjne was consistently low across most of the study area (Figure 1b), whereas TDβjtu displayed generally higher values across many watersheds (Figure 1c). TDβjac was predominantly shaped by turnover (mean: 0.25 ± 0.13), with nestedness playing only a minor role (mean: 0.18 ± 0.09) (Figure S2). FDβjac displayed a spatially heterogeneous pattern across the study area, with higher values predominantly clustered in the western region (Figure 1d). Over half of the watersheds (51%) had FDβjac values exceeding 0.5, indicating widespread functional differentiation in these areas. In contrast, PDβjac exhibited elevated values mainly in the central and eastern regions (Figure 1g). Most watersheds (96.4%) showed PDβjac values below 0.5, reflecting more limited phylogenetic divergence across most of the study watersheds. Decomposition analyses revealed that the nestedness component contributed substantially more than turnover to both FDβjac and PDβjac (Figure S2). Specifically, FDβjne (mean ± SD: 0.34 ± 0.16) accounted for the majority of FDβjac (0.52 ± 0.19), while PDβjne (0.17 ± 0.09) also played a dominant role in total PDβjac (0.30 ± 0.11). Spatially, higher values of both FDβjne and the FDβjtu were concentrated in the west (Figure 1e,f), whereas PDβjne and PDβjtu displayed broadly similar geographic distributions across the entire study area (Figure 1h,i). Among the three dimensions of beta diversity, FDβjac exhibited the highest values. A notable contrast emerged in the partitioning results: while taxonomic beta diversity was dominated by the turnover component, the nestedness component showed a greater contribution to both functional and phylogenetic dimensions (Table S2).

3.2 | Correlation Analysis of Three Facets of Beta Diversity and Their Components

Spearman correlation analysis revealed that the three dimensions of beta diversity and their components exhibited generally strong and significant positive spatial correlations across watersheds (Figure 2). The correlation coefficients (R) ranged from 0.46 (FDβjac and PDβjac) to 0.77 (TDβjtu and PDβjtu). Overall, TDβjac and PDβjac showed the highest spatial congruence, whereas FDβjac displayed relatively distinct distribution patterns, with consistently weaker correlations to both TDβjac and PDβjac (Figure 2). This pattern was also evident in the decomposed components, as both nestedness and turnover correlated most strongly between the taxonomic and phylogenetic dimensions. Notably, correlations among nestedness components were generally stronger than those of total beta diversity or turnover components. This high congruence between taxonomic and phylogenetic diversity suggests a pronounced phylogenetic signal and shared evolutionary constraints, while the moderate correlation between functional diversity and the other two dimensions reflects the combined effects of environmental filtering and adaptive differentiation in shaping heterogeneous community patterns at the trait level.

3.3 | Effects of Environmental Drivers on Beta Diversity

The BRT models revealed substantial but varying explanatory power for different dimensions of freshwater fish diversity (Figures 3–5). Contemporary precipitation category was the most consistently key driver for TDβjac and FDβjac, whereas climate anomaly category had the most influence for PDβjac.

For TDβjac, the model explained 33.4% of the total deviance (Figure 3). Contemporary precipitation factors were the most influential predictor group (collectively contributing 30.3%). At the individual factor level, Velocity (a climate anomaly factor) emerged as the primary contributor (14.7%), highlighting the role of past climate shifts. This was closely followed by Distance (12%), which underscores the significant influence of geographic isolation

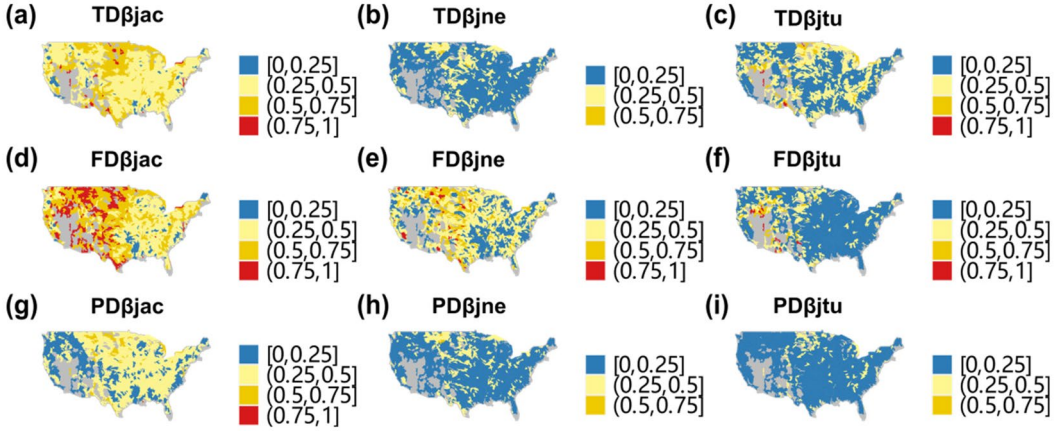


FIGURE 1 | Maps of freshwater fishes taxonomic (TDβjac, TDβjne and TDβjtu), functional (FDβjac, FDβjne and FDβjtu) and phylogenetic beta diversity phenomena (PDβjac, PDβjne and PDβjtu) in the US watersheds. FDβjac, total functional diversity; FDβjne, nestedness functional diversity; FDβjtu, turnover functional diversity; PDβjac, total phylogenetic diversity; PDβjne, nestedness phylogenetic diversity; PDβjtu, turnover phylogenetic diversity; TDβjac, total taxonomic diversity; TDβjne, nestedness taxonomic diversity; TDβjtu, turnover taxonomic diversity.

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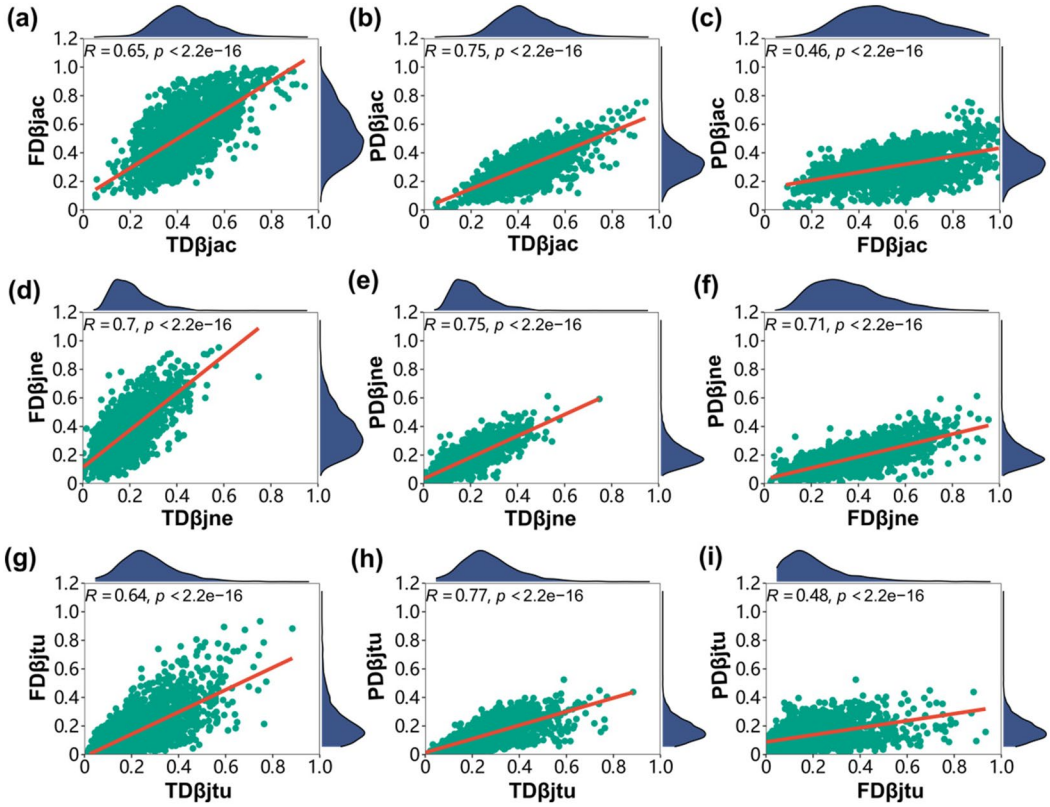


FIGURE 2 | Correlation plots of fishes taxonomic (TDβjac, TDβjne and TDβjtu), functional (FDβjac, FDβjne and FDβjtu) and phylogenetic (PDβjac, PDβjne and PDβjtu) beta diversity phenomena across the US watersheds. Correlation coefficients (R) are shown for each relationship based on Spearman test. FDβjac, total functional diversity; FDβjne, nestedness functional diversity; FDβjtu, turnover functional diversity; PDβjac, total phylogenetic diversity; PDβjne, nestedness phylogenetic diversity; PDβjtu, turnover phylogenetic diversity; TDβjac, total taxonomic diversity; TDβjne, nestedness taxonomic diversity; TDβjtu, turnover taxonomic diversity.

and inter-watershed connectivity on shaping TDβjac patterns (Figure 3). Predictors explained 16.8% of nestedness (TDβjne) and 28.9% of turnover (TDβjtu), with Velocity and Distance being key for TDβjne and TDβjtu, respectively (Figures S3 and S4). For FDβjac, the model accounted for 46.1% of the deviance. Contemporary precipitation factors again formed the most influential group (33.7%). At the individual level, Elevation (13.9%) and Runoff (11.9%) were top contributors, exhibiting similar nonlinear trends (Figure 4). Predictors explained 19.7% of FDβjne (led by Runoff) and 31.5% of FDβjtu (dominated by Distance) (Figures S5 and S6). In contrast, for PDβjac, the model explained 33.5% of the total deviance, with climate anomaly factors constituting the most influential group (37.4%). Velocity was the dominant single factor (26.4%), showing a positive relationship with PDβjac, while Distance also remained significant (Figure 5). Predictors explained 17.1% of PDβjne (led by Velocity) and 25.8% of PDβjtu (also led by Velocity) (Figures S7 and S8). Regression analysis on these nine metrics revealed that the relationships between predictors and diversity were predominantly complex and nonlinear. The primary drivers diverged somewhat across different diversity dimensions, yet the effects of most influential factors aligned in their trends.

4 | Discussion

Studying beta diversity from various perspectives is pivotal for comprehensively analysing patterns and the factors influencing biodiversity (Villéger et al. 2013; Gianuca et al. 2017). Consequently, we conducted an analysis of taxonomic, functional and phylogenetic beta diversity patterns, examining their correlations with abotic factors that may act as determinants. This investigation utilised a comprehensive dataset encompassing native freshwater fish species across the US.

4.1 | Multiplefaceted Beta Diversity Patterns and Their Relationship

The results revealed that FDβjac surpassed both TDβjac and PDβjac (based on average values). This pattern suggests that differences in environmental conditions (such as elevation) between watersheds exert a stronger filtering effect on species' functional traits than on their taxonomic identities or evolutionary lineages. The high FDβjac values reflect marked functional differentiation among watersheds. This pattern may be driven by niche differentiation in response to heterogeneous resource availability resulting from environmental gradients such as elevation and runoff, thereby enabling species to adapt to locally distinct ecological processes (Cantera et al. 2023). Furthermore, due to environmental constraints, connectivity between populations is relatively low. Correlation analyses indicate a significant relationship between taxonomic and functional diversity (Figure 2), suggesting that limited population connectivity may promote the distribution of species with different traits across watersheds, thereby generating the observed pattern. This result aligns with other findings in fish diversity research (Wu, Lv,

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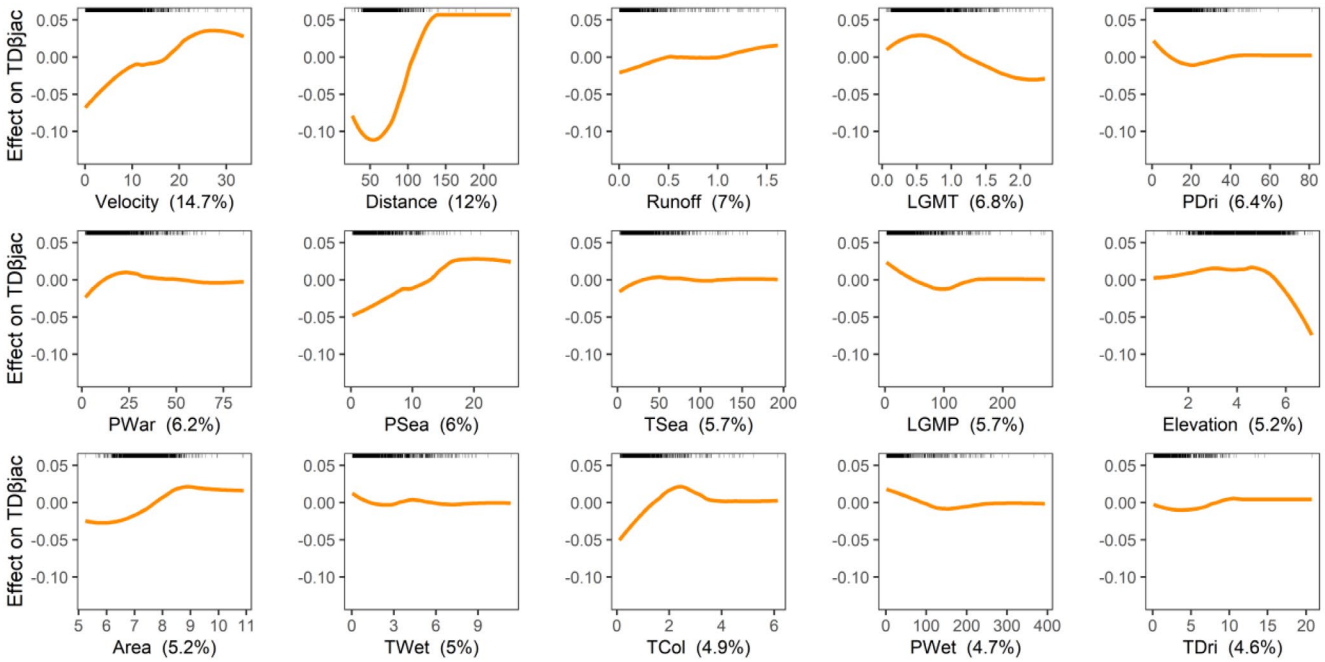


FIGURE 3 | Results of boosted regression trees showing the partial dependency between taxonomic total beta diversity (TDβjac) and predictors. The value in parentheses in each panel shows the percentage of contribution of each predictor considered in the model. The rugs in the top of each panel show the distribution of the watersheds along the predictor values. Area, watershed area; Distance, mean geographic distance to the five nearest neighbouring watersheds; LGMT & LGMP, change in mean annual temperature and precipitation between present and Last Glacial Maximum conditions; PDri, precipitation of driest quarter; PSea, precipitation seasonality; PWar, precipitation of warmest quarter; PWet, precipitation of wettest quarter; TCol, mean temperature of coldest quarter; TDri, mean temperature of driest quarter; TSea, temperature seasonality; TWet, mean temperature of wettest quarter; Velocity, Late Quaternary glacial–interglacial climate-change velocity.

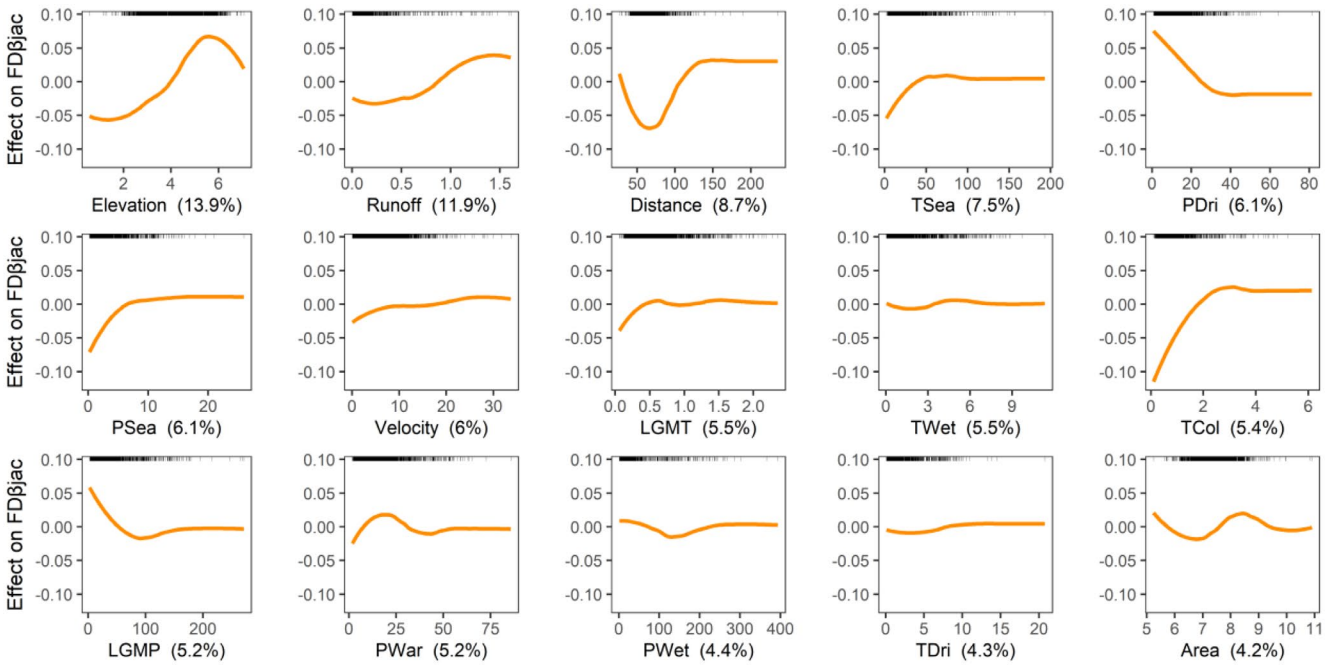


FIGURE 4 | Results of boosted regression trees showing the partial dependency between functional total beta diversity (FDβjac) and predictors. The value in parentheses in each panel shows the percentage of contribution of each predictor considered in the model. The rugs in the top of each panel show the distribution of the watersheds along the predictor values. Area, watershed area; Distance, mean geographic distance to the five nearest neighbouring watersheds; LGMT & LGMP, change in mean annual temperature and precipitation between present and Last Glacial Maximum conditions; PDri, precipitation of driest quarter; PSea, precipitation seasonality; PWar, precipitation of warmest quarter; PWet, precipitation of wettest quarter; TCol, mean temperature of coldest quarter; TDri, mean temperature of driest quarter; TSea, temperature seasonality; TWet, mean temperature of wettest quarter; Velocity, Late Quaternary glacial–interglacial climate-change velocity.

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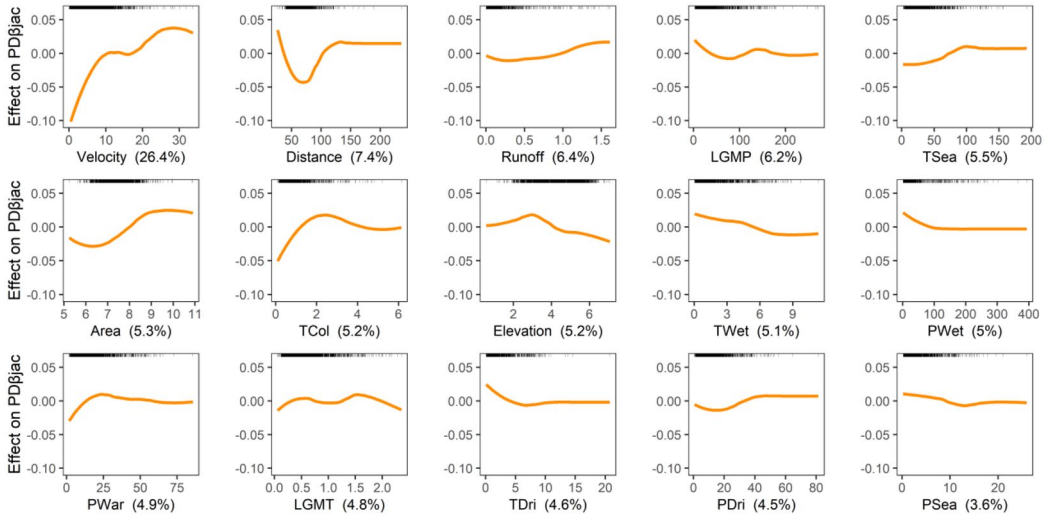


FIGURE 5 | Results of boosted regression trees showing the partial dependency between phylogenetic total beta diversity (PDβjac) and predictors. The value in parentheses in each panel shows the percentage of contribution of each predictor considered in the model. The rugs in the top of each panel show the distribution of the watersheds along the predictor values. Area, watershed area; Distance, mean geographic distance to the five nearest neighbouring watersheds; LGMT & LGMP, change in mean annual temperature and precipitation between present and Last Glacial Maximum conditions; PDri, precipitation of driest quarter; PSea, precipitation seasonality; PWar, precipitation of warmest quarter; PWet, precipitation of wettest quarter; TCol, mean temperature of coldest quarter; TDri, mean temperature of driest quarter; TSea, temperature seasonality; TWet, mean temperature of wettest quarter; Velocity, Late Quaternary glacial–interglacial climate-change velocity.

et al. 2022; Hernández-Mendoza et al. 2024). When comparing TDβ and PDβ, TDβ exhibited higher values than PDβ, which suggests that while species dissimilarity is high, PDβ captures a deeper level of evolutionary history similarity for some different species (Podani et al. 2018). In other words, this suggests that some species dissimilarity may be driven by shared evolutionary history, such as the replacement of congeneric species (Zhang et al. 2025). The results also indicate that patterns of TDβ and PDβ may be jointly influenced by environmental filtering and phylogenetic conservatism (Geheber 2019; Zhang et al. 2025). These findings imply that conservation practices should take into account the evolutionary history of species to avoid overestimating phylogenetic differences among communities.

Species turnover emerged as the dominant component of TDβjac, indicating that differences in species composition among watersheds are primarily driven by species replacement rather than species loss. Watersheds with high species turnover are often associated with environmental heterogeneity, where factors such as Late Quaternary glacial–interglacial climate-change velocity and runoff exert strong environmental filtering effects, promoting differentiation in species composition (Chen et al. 2025). Our results show that greater differences in climate-change velocity and runoff between adjacent watersheds reduce the number of shared species and thus increase taxonomic turnover. In addition, unlike more mobile taxa (e.g., birds or mammals), freshwater fishes are notably constrained by physical landscape features (such as distance factor) (Radinger et al. 2018). Due to the limited connectivity among river basins, fish assemblages exhibit strong differentiation, largely driven by high species turnover, as previously reported and confirmed by our TDβjac analysis (Leprieur et al. 2011).

In contrast, both functional beta diversity (FDβjac) and phylogenetic beta diversity (PDβjac) were dominated by nestedness, partially supporting our first hypothesis. At the functional level, differences in elevation and runoff between adjacent watersheds intensify environmental filtering, reducing functional diversity and disproportionately affecting species with rare trait combinations or those belonging to particular phylogenetic lineages (Pease et al. 2012). Consequently, communities in environmentally harsher watersheds tend to represent functional subsets of neighbouring communities, generating a nested pattern in functional diversity (Wu et al. 2021; Wu, Wang, et al. 2022).

A similar mechanism appears to underlie phylogenetic nestedness. The nested pattern in phylogenetic diversity may arise because phylogenetically older lineages generally possess greater environmental adaptability (e.g., to Late Quaternary glacial–interglacial climate-change velocity), whereas lineages with higher evolutionary distinctiveness tend to exhibit more restricted distribution ranges (Winter et al. 2013). The significant influence of climate-change velocity on PDβjac supports this interpretation. In regions experiencing greater climatic instability, strong environmental filtering likely eliminates lineages with lower adaptive capacity, while evolutionarily distinct and range-restricted species persist mainly in climatically stable refugia. The retention of these regional differences reinforces nested patterns in phylogenetic composition. Overall, the observed patterns of freshwater fish beta diversity, together with the contrasting contributions of turnover and nestedness, are consistent with findings reported at broader spatial scales and support general ecological principles governing large-scale biodiversity patterns (Chen et al. 2025).

The results demonstrate significant positive correlations among all three beta diversity indices and their components. These findings support our second hypothesis, indicating a high degree of congruence in beta diversity across taxonomic, functional and phylogenetic dimensions. Although

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taxonomic diversity was higher than phylogenetic diversity in this study, the correlation between the two exceeded that observed among other diversity dimensions. This strong correlation suggests that fish speciation is shaped by consistent evolutionary mechanisms and shared ancestral lineages (Xia et al. 2023; Chen et al. 2025). Consequently, taxonomic and phylogenetic diversity respond congruently to environmental drivers, such as Late Quaternary climate velocity and geographic distance. Besides, this observed synergy could be further methodologically reinforced, as inferring phylogenetic placements for certain species based on taxonomic information inherently biases the two datasets towards greater alignment (Rabosky et al. 2018). However, compared to the strong correlation between TDβjac and PDβjac, the associations of FDβjac with both TDβjac and PDβjac were notably weaker, with the FDβjac–PDβjac correlation being the weakest. This pattern highlights a relative decoupling between functional structure and both taxonomic and phylogenetic composition. By integrating environmental factors, this pattern suggests that under comparable Late Quaternary glacial–interglacial climate-change velocities, distantly related species may converge within a functional trait space, thereby driving functional convergence at the community level. In contrast, within heterogeneous environments characterised by significant topographic variation, closely related species tend to exhibit greater functional differentiation—likely an adaptive response to localised environmental pressures (Lin et al. 2021; Hernández-Mendoza et al. 2024).

4.2 | Determinants of Beta Diversity Patterns

Identifying the geographic and climatic drivers of biotic change is essential for understanding biodiversity and developing effective management strategies (Socolar et al. 2016; Teichert et al. 2018). This study demonstrates that contemporary climate and climatic anomalies are significantly associated with TDβjac, FDβjac and PDβjac, partially supporting our third hypothesis. Notably, velocity, elevation, runoff and distance emerged as the most influential factors shaping diversity patterns.

As a key predictor of taxonomic and phylogenetic diversity, climate change velocity reflects how rapid Quaternary climatic shifts often trigger abrupt environmental transformations. The widening disparity in Late Quaternary glacial–interglacial climate-change velocity among different watersheds amplifies the spatial divergence of environmental conditions: watersheds with higher velocities experience more intense environmental fluctuations, thereby elevating species extinction risk; by contrast, watersheds characterised by relatively low Quaternary climatic velocities exhibit more stable environmental conditions, which favour the persistence of species (Griffiths 2010, 2012; Sandel et al. 2011; Comte and Grenouillet 2015; Buckley et al. 2024). There exists a close relationship between species composition and Late Quaternary glacial–interglacial climate-change velocity, with endemic species being particularly vulnerable to local extinction due to environmental filtering (Sandel et al. 2011). Furthermore, watersheds with varying Late Quaternary glacial–interglacial climate-change velocity selectively filter lineages with different climatic tolerances, thereby intensifying the influence of inter-watershed differences on taxonomic and phylogenetic diversity—the greater the disparity, the more pronounced its shaping effect on diversity (Leprieur et al. 2011; Dobrovolski et al. 2012).

Elevation differences are an important driver of spatial variation in FDβjac. The spatial distribution results of this study show that in the central region, where elevations are relatively low, environmental conditions among watersheds are relatively similar, resulting in lower FDβjac values; in contrast, the high-elevation mountainous areas in the west exhibit complex terrain and higher FDβjac values, which are primarily driven by the nestedness component. Previous studies have shown that topographic complexity and geographic isolation are important factors shaping spatial patterns of β-diversity (Jaramillo-Villa et al. 2010; Bhatt et al. 2013; Cheng et al. 2019). The BRT analysis in this study further indicates that FDβjac increases with greater inter-watershed elevation differences. Larger elevation differences are generally associated with stronger environmental filtering and lower landscape connectivity, thereby limiting species dispersal (Pease et al. 2012; Dodds et al. 2019). In this process, only species with adaptive functional trait combinations are able to persist, while some functional traits are lost in local watersheds (trait loss) (Raffard et al. 2020). As a result, some watersheds retain only a subset of the functional spectrum of other watersheds rather than forming completely distinct functional compositions, leading to increases in FDβjac that are primarily driven by the nestedness component rather than functional turnover (Lamouroux et al. 2002; Xia et al. 2024).

As the difference in runoff between watersheds increases, all three diversity indices show an upward trend, with the increase in FDβjac being particularly pronounced. Runoff often serves as a proxy for habitat size; substantial differences in runoff between watersheds typically reflect variations in river scale. Indeed, the transition from small streams to large tributaries leads to shifts in fish fauna composition (Oberdorff et al. 2011). Furthermore, distinct runoff characteristics shape divergent habitat conditions, which in turn select for species adapted to specific hydrological regimes through environmental filtering. This process reduces species overlap and drives the differentiation of functional trait combinations between watersheds (Gutiérrez et al. 2018).

In this study, we incorporated geographic distance between watersheds as a key spatial factor in our models and found that it significantly influences the formation of multi-dimensional biodiversity patterns. At relatively short distances, geographic barriers are minimal, facilitating frequent species dispersal between neighbouring watersheds, which leads to homogenisation of community composition (Heino et al. 2015; Pompeu et al. 2023). As geographic distance among watersheds increases, the cumulative effects of dispersal limitation gradually intensify, leading to significant differentiation in taxonomic, functional and phylogenetic composition between watersheds (Heino et al. 2015; Jiang et al. 2024). Our results indicate that geographic distance is not only a common dominant predictor of all three facets of total beta diversity (TDβjac, FDβjac, PDβjac) but also the strongest explanatory variable for the TDβjtu and FDβjtu. Further analysis of TDβjtu, FDβjtu and PDβjtu reveals that, across all three dimensions of diversity, the influence of spatial isolation on community differentiation follows a nonlinear pattern.

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Specifically, as geographic distance increases, its facilitative effect on differentiation gradually weakens and tends towards an asymptote at larger distances; beyond this threshold, additional increases in distance contribute little to further biodiversity differentiation (Figures S4, S6 and S8). This 'saturation effect' highlights the pivotal role of dispersal limitation as a core driver of multidimensional biodiversity patterns, while also indicating that spatially driven differentiation has an inherent upper limit.

4.3 | Conservation Implications

The beta diversity patterns revealed in this study indicate that watersheds with high beta diversity are primarily located in the central and western regions of the US, particularly with respect to FDβ, highlighting the conservation priority of these areas (Margules and Pressey 2000; Barboza et al. 2015). Importantly, different dimensions of beta diversity are shaped by distinct ecological processes. Taxonomic beta diversity is dominated by species turnover, implying that most watersheds contain unique species assemblages. This highlights the need for broad, multi-watershed conservation strategies (Rodríguez-Lozano et al. 2023), including the expansion of protected areas in high-elevation regions of western US and the preservation of spatial heterogeneity among watersheds to ensure that unique biodiversity components are effectively represented (Leprieur et al. 2011; Luiz et al. 2025). In contrast, functional and phylogenetic beta diversity are mainly structured by nestedness, indicating that conservation efforts should prioritise watersheds with the highest functional and phylogenetic richness. Because nested patterns suggest that species-poor watersheds are subsets of richer ones, safeguarding these core high-richness areas can maximise regional conservation outcomes (Baselga 2010; Leprieur et al. 2011). Overall, the discrepancies between nestedness- and turnover-driven patterns across different diversity dimensions emphasise that future fish conservation must adopt a holistic, metacommunity-based strategy rather than relying on single-metric approaches (Xu et al. 2015).

This study advances the understanding of freshwater fish biodiversity by revealing the spatial patterns and driving mechanisms of taxonomic, functional and phylogenetic beta diversity. The findings emphasise that future conservation strategies should adopt a multi-faceted perspective, prioritising both the protection of species replacement networks and the conservation of high-diversity core areas.

5 | Conclusion

To conclude, identifying the predominant factors influencing beta diversity is crucial for informing policymakers about effective conservation strategies. This study analyses the beta diversity patterns of freshwater fishes in the US from taxonomic, functional and phylogenetic dimensions. The results indicate that environmental filtering and dispersal limitation jointly shape the diversity characteristics across different dimensions. Taxonomic diversity is dominated by species turnover, while functional and phylogenetic diversity exhibit a pronounced nested structure. The spatial patterns of the three diversity dimensions are consistent, further supporting the intrinsic correlations among multidimensional beta diversity. Late Quaternary glacial–interglacial climate-change velocity, elevation and runoff are key environmental factors driving diversity patterns, while geographic distance nonlinearly modulates community composition differences through dispersal limitation. Overall, our results emphasise the importance of considering multiple facets of beta diversity in the analysis and development of conservation plans.

Author Contributions

Congjun Xu, **Naisen Liu**, **Jun Xu** and **Guohuan Su:** conceptualisation. **Congjun Xu**, **Chenggong Du**, **Naisen Liu**, **Taotao Dai** and **Guohuan Su:** methodology. **Congjun Xu**, **Huiru Xue**, **Jiping Li**, **Yingxin Li**, **Chenggong Du**, **Wei Zhang** and **Guohuan Su:** formal analysis. **Congjun Xu:** writing – original draft and visualisation. **Huiru Xue** and **Guohuan Su:** data curation. **Jun Xu** and **Guohuan Su:** supervision. **Jiping Li** and **Guohuan Su:** funding acquisition. All author: Writing – review and editing.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data openly available in a public repository that issues datasets with DOIs: The data that support the findings of this study are openly available in [Zenodo] at https://doi.org/10.5281/zenodo.17208070 and https://doi.org/10.5281/zenodo.17939831.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** Correlation analysis between the 15 predictors. Area: watershed area; Velocity: Late Quaternary glacial–interglacial climate-change velocity; LGMT & LGMP: change in mean annual temperature and precipitation between present and Last Glacial Maximum conditions; TSea: temperature seasonality; PSea: precipitation seasonality; TWet: mean temperature of wettest quarter; TDri: mean temperature of driest quarter; TCol: mean temperature of coldest quarter; PWet: precipitation of wettest quarter; PDri: precipitation of driest quarter; PWar: precipitation of warmest quarter; Distance: mean geographic distance to the five nearest neighbouring watersheds. **Figure S2:** Maps of freshwater fishes turnover and nestedness ratio (turnover or nestedness/total) for taxonomic (TDβjne_r and TDβjtu_r), functional (FDβjne_r and FDβjtu_r) and phylogenetic (PDβjne_r and PDβjtu_r) beta diversity phenomena in the US watersheds. TDβjne_r: nestedness taxonomic diversity ratio; TDβjtu_r: turnover taxonomic diversity ratio; FDβjne_r: nestedness functional diversity ratio; FDβjtu_r: turnover functional diversity ratio; PDβjne_r: nestedness phylogenetic diversity ratio; PDβjtu_r: turnover phylogenetic diversity ratio. **Figure**

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S3: Results of boosted regression trees showing the partial dependency between taxonomic nestedness beta diversity (TDβjne) and 15 predictors. The values in parentheses in each panel shows the percentage of contribution of each predictor considered in the model. The rugs in the top of each panel show the distribution of the watersheds along the predictor values. Area: watershed area; Velocity: Late Quaternary glacial–interglacial climate-change velocity; LGMT & LGMP: change in mean annual temperature and precipitation between present and Last Glacial Maximum conditions; TSea: temperature seasonality; PSea: precipitation seasonality; TWet: mean temperature of wettest quarter; TDri: mean temperature of driest quarter; TCol: mean temperature of coldest quarter; PWet: precipitation of wettest quarter; PDri: precipitation of driest quarter; PWar: precipitation of warmest quarter; Distance: mean geographic distance to the five nearest neighbouring watersheds.
Figure S4: Results of boosted regression trees showing the partial dependency between taxonomic turnover beta diversity (TDβjtu) and 15 predictors. The values in parentheses in each panel shows the percentage of contribution of each predictor considered in the model. The rugs in the top of each panel show the distribution of the watersheds along the predictor values. Area: watershed area; Velocity: Late Quaternary glacial–interglacial climate-change velocity; LGMT & LGMP: change in mean annual temperature and precipitation between present and Last Glacial Maximum conditions; TSea: temperature seasonality; PSea: precipitation seasonality; TWet: mean temperature of wettest quarter; TDri: mean temperature of driest quarter; TCol: mean temperature of coldest quarter; PWet: precipitation of wettest quarter; PDri: precipitation of driest quarter; PWar: precipitation of warmest quarter; Distance: mean geographic distance to the five nearest neighbouring watersheds.
Figure S5: Results of boosted regression trees showing the partial dependency between functional nestedness beta diversity (FDβjne) and 15 predictors. The values in parentheses in each panel shows the percentage of contribution of each predictor considered in the model. The rugs in the top of each panel show the distribution of the watersheds along the predictor values. Area: watershed area; Velocity: Late Quaternary glacial–interglacial climate-change velocity; LGMT & LGMP: change in mean annual temperature and precipitation between present and Last Glacial Maximum conditions; TSea: temperature seasonality; PSea: precipitation seasonality; TWet: mean temperature of wettest quarter; TDri: mean temperature of driest quarter; TCol: mean temperature of coldest quarter; PWet: precipitation of wettest quarter; PDri: precipitation of driest quarter; PWar: precipitation of warmest quarter; Distance: mean geographic distance to the five nearest neighbouring watersheds.
Figure S6: Results of boosted regression trees showing the partial dependency between functional turnover beta diversity (FDβjtu) and 15 predictors. The values in parentheses in each panel shows the percentage of contribution of each predictor considered in the model. The rugs in the top of each panel show the distribution of the watersheds along the predictor values. Area: watershed area; Velocity: Late Quaternary glacial–interglacial climate-change velocity; LGMT & LGMP: change in mean annual temperature and precipitation between present and Last Glacial Maximum conditions; TSea: temperature seasonality; PSea: precipitation seasonality; TWet: mean temperature of wettest quarter; TDri: mean temperature of driest quarter; TCol: mean temperature of coldest quarter; PWet: precipitation of wettest quarter; PDri: precipitation of driest quarter; PWar: precipitation of warmest quarter; Distance: mean geographic distance to the five nearest neighbouring watersheds.
Figure S7: Results of boosted regression trees showing the partial dependency between phylogenetic nestedness beta diversity (PDβjne) and 15 predictors. The values in parentheses in each panel shows the percentage of contribution of each predictor considered in the model. The rugs in the top of each panel show the distribution of the watersheds along the predictor values. Area: watershed area; Velocity: Late Quaternary glacial–interglacial climate-change velocity; LGMT & LGMP: change in mean annual temperature and precipitation between present and Last Glacial Maximum conditions; TSea: temperature seasonality; PSea: precipitation seasonality; TWet: mean temperature of wettest quarter; TDri: mean temperature of driest quarter; TCol: mean temperature of coldest quarter; PWet: precipitation of wettest quarter; PDri: precipitation of driest quarter; PWar: precipitation of warmest quarter; Distance: mean geographic distance to the five nearest neighbouring watersheds.
Figure S8: Results of boosted regression trees showing the partial dependency between phylogenetic turnover beta diversity (PDβjtu) and 15 predictors. The values in parentheses in each panel shows the percentage of contribution of each predictor considered in the model. The rugs in the top of each panel show the distribution of the watersheds along the predictor values. Area: watershed area; Velocity: Late Quaternary glacial–interglacial climate-change velocity; LGMT & LGMP: change in mean annual temperature and precipitation between present and Last Glacial Maximum conditions; TSea: temperature seasonality; PSea: precipitation seasonality; TWet: mean temperature of wettest quarter; TDri: mean temperature of driest quarter; TCol: mean temperature of coldest quarter; PWet: precipitation of wettest quarter; PDri: precipitation of driest quarter; PWar: precipitation of warmest quarter; Distance: mean geographic distance to the five nearest neighbouring watersheds.
Table S1: 24 Predictor variables considered in the analysis, their abbreviation and reference sources. **Table S2:** Beta diversity and its components of freshwater fishes in the US.

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